From Shortest Route to Smartest Business Route

LogiPilot: AI-Powered Enterprise Logistics Decision Intelligence

> Because the shortest route can still be the wrong business decision.

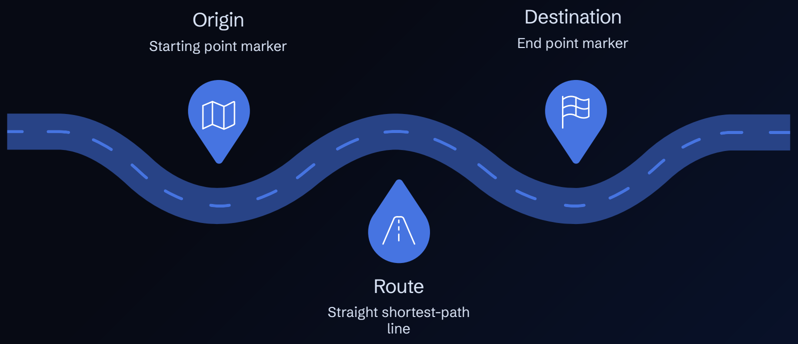
Agent Forge · TCS AI Friday

THE PROBLEM

The Route Is More Than a Line on a Map

Conventional Routing

Simple shortest-path logic on a monochrome map.



Enterprise Decision Problem

Real-Time Disruptions

Traffic, weather, incidents change continuously.

Business Constraints

Cost, SLA, fuel, tolls, priorities.

Cargo Sensitivity

Medicine, perishables, hazmat requirements.

Vehicle & Driver Limits

Weight, clearance, HOS compliance.

Fragmented Data

APIs, systems, operational sources.

Traditional shortest-path routing optimizes distance/time — not the complete business decision.

OUR PLATFORM

From Shortest Path to Business-Optimal Path

A transformation flow that shows how raw routing inputs become a business-aware recommendation.

INPUT

Shipment + Vehicle + Driver + Business Constraints + Real-Time Data

MULTI-AGENT LANGGRAPH ORCHESTRATION

Intent Parser | Constraint Agent | Graph Retrieval | Traffic Agent

ML DELAY / RISK PREDICTION

Predict delays, route risk, ETA

DYNAMIC ENTERPRISE COST MODEL

Combine time, distance, cost, product, vehicle, driver, SLA, traffic, weather, risk

A* MULTI-OBJECTIVE ROUTE OPTIMIZATION

Find lowest-cost feasible path

Shortest Route

120 km | ₹X | 75 min | High Risk

Smart Route

128 km | ₹Y | 68 min | Low Risk

TIME · DISTANCE · COST · PRODUCT · VEHICLE · DRIVER HOS · SLA · TRAFFIC · WEATHER · RISK

Subject to operational & regulatory constraints

We optimize the decision — not just the distance.

CONSTRAINT INTELLIGENCE

Enterprise Constraints Drive the Decision

Different route constraints compete and combine to determine the best business outcome.

TIME

- ETA
- Delay
- SLA
- Driver HOS

COST

- Fuel
- Toll
- Operational Cost

CARGO

- Product Type
- Priority
- Cold Chain
- Hazmat Requirements

VEHICLE

- Vehicle Type
- Capacity
- Weight
- Clearance

ENVIRONMENT

- Traffic
- Weather
- Incidents
- Road Risk

These constraints become inputs to our Dynamic Cost Model and Constraint Engine.

Different shipments → different optimization priorities

MEDICINE

Time + Safety + Cold Chain

FURNITURE

Cost + Distance + Road Quality

EMERGENCY

Time + Reliability

CONSTRAINT INTELLIGENCE

How LogiPilot Thinks

An end-to-end intelligence pipeline that combines AI, ML, and deterministic optimization

DATA
Real-time APIs + Enterprise Data + Shipment Data

INTENT PARSER AGENT
Parse user request, extract constraints

CONSTRAINT AGENT
Validate business constraints

GRAPH RETRIEVAL AGENT
Retrieve topology & logistics metadata from Neo4j

TRAFFIC AGENT
Live traffic data from Google Routes API

DELAY PREDICTION AGENT
ML-based delay & risk prediction

PATHFINDING AGENT
A* + Yen's K-Shortest Paths

FEASIBILITY AGENT
Validate routes against constraints

RECOMMENDATION AGENT
Select optimal route

ROOT CAUSE AGENT
Explain decision rationale

REFLECTION AGENT
Self-correction loop

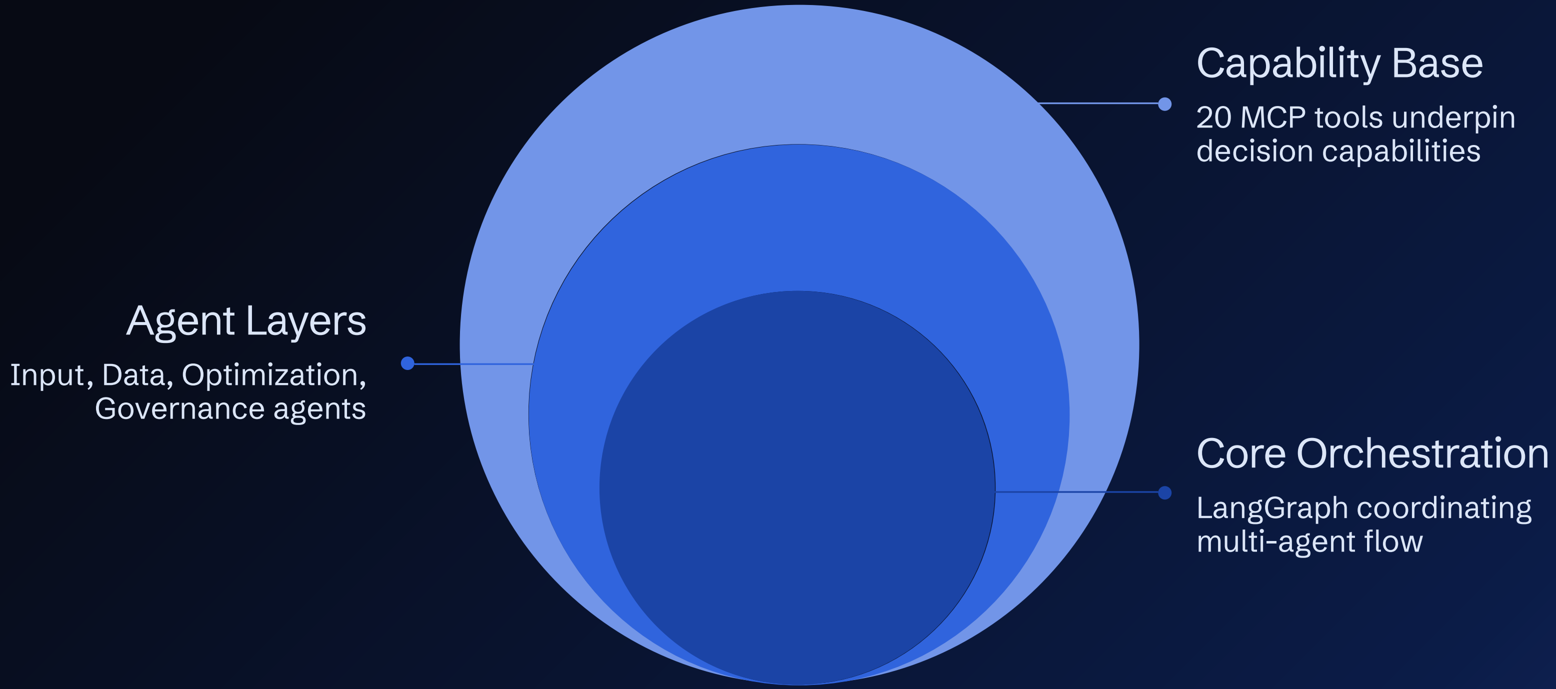
REPORT AGENT
Generate output & recommendations

LangGraph Multi-Agent Orchestration with Reflection Loop

ML + MULTI-AGENT INTELLIGENCE

12 Specialized Agents. One Coordinated Decision.

LangGraph multi-agent orchestration with reflection and self-correction



Each agent has a focused responsibility — LangGraph coordinates the overall decision.

> 20 MCP tools provide controlled access to external capabilities.

SYSTEM ARCHITECTURE

Where AI Meets Optimization

From real-time data to business-optimal route decisions

01

REAL-TIME DATA

Traffic | Weather | GPS | Incidents

02

SHIPMENT CONTEXT

Product | Priority | SLA

03

VEHICLE & DRIVER

Vehicle Type | Capacity | HOS

04

ML DELAY & RISK PREDICTION

Expected Delay
ETA
Route Risk

05

DYNAMIC EDGE COST

Time | Distance | Cost | Traffic | Weather | Product Priority | Vehicle Constraints | Driver HOS | SLA | Risk | CO_2

06

A* SEARCH

Find the lowest-cost feasible path

07

YEN'S K-SHORTEST PATH

Generate distinct alternatives

08

CONSTRAINT-AWARE FILTER

Reject routes violating hard constraints

09

FINAL BUSINESS-OPTIMAL ROUTE

ML predicts what may happen. Optimization decides what to do.

A* does not optimize distance alone.
It optimizes our dynamic enterprise cost.

$$RouteCost = Time + Distance + BusinessCost + Risk + ConstraintPenalties$$

SYSTEM ARCHITECTURE

LogiPilot AI — System Architecture

Multi-agent AI orchestration with deterministic optimization and Neo4j knowledge integration

USER EXPERIENCE
Ops Manager · Driver · Operations Dashboard · Driver Mobile / Vehicle Console · React 19 + Vite 6

EXTERNAL SERVICES
OpenAI-compatible LLM Gateway · Google Routes API (live traffic) · Open-Meteo API (weather) · DuckDuckGo Web Search · LangSmith (tracing & observability)

BACKEND
FastAPI · REST API / SSE Streaming

12-AGENT LANGGRAPH WORKFLOW
Intent Parser Agent · Constraint Agent · Graph Retrieval Agent · Pathfinding Agent · Traffic Agent · Delay Prediction Agent · Feasibility Agent · Recommendation Agent · Root Cause Agent · Reflection Agent · Learning Agent · Report Agent

ROUTING ENGINE
A* Search · Yen's K-Shortest Paths · Constraint-Aware Filter

MCP TOOL LAYER
Network & Graph (4) · Traffic & Live Data (4) · Weather & Geo (4) · Constraints & Rules (4) · Analytics & Insights (4)

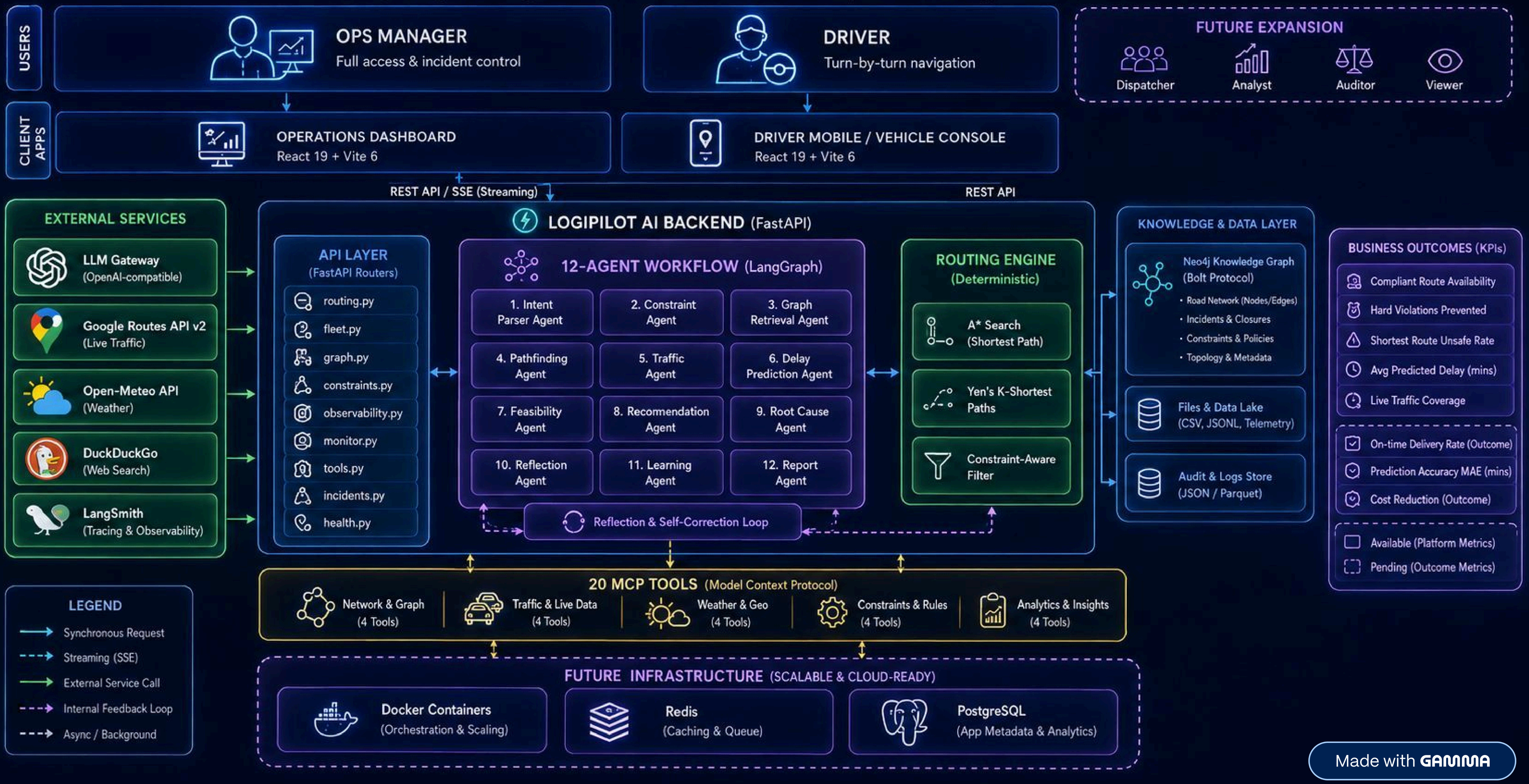
KNOWLEDGE & DATA
Neo4j Knowledge Graph (topology, constraints, policies, metadata) · Files / Data Lake · Audit & Logs Store

FUTURE INFRASTRUCTURE
Docker · Redis · PostgreSQL

Business outcomes: Improved on-time delivery · Reduced violations · Better route utilization · Improved delay prediction · Reduced cost · Improved operational visibility

LOGIPILOT AI — SYSTEM ARCHITECTURE

Multi-agent Logistics Intelligence Platform for Constraint-Aware Route Optimization & Delay Prediction



DIFFERENTIATION

One Shipment. One Decision. Multiple Constraints.

Shipment: Medicine — Critical Priority

Route: Kochi → Thiruvananthapuram

LIVE INPUTS

Traffic · Weather · Incidents

BUSINESS CONSTRAINTS

Cold Chain · SLA · Vehicle suitability · Driver HOS · Cost

AGENT EXECUTION SEQUENCE

01

Dispatcher creates shipment

02

Intent + Constraint Agents
Parse request, validate constraints

03

Traffic / Weather / Graph Agents
Retrieve live data and topology

04

Delay Prediction Agent
Predicts delay + ETA + risk

05

Optimization Agent
Dynamic Cost + A*

06

Yen's K-Shortest Paths
Generate alternative routes

07

Feasibility Agent
Constraint validation

08

Recommendation Agent
Select optimal route

09

Driver receives recommended route

ENTERPRISE AI LOGISTICS

From Logistics Optimization to Enterprise Decision Intelligence

BUSINESS OUTCOMES

FASTER
Predictive delay management

CHEAPER
Cost-aware route optimization

SAFER
Constraint-aware routing

SMARTER
Multi-agent explainable decisions

ROADMAP PROGRESSION

TODAY
Logistics Route Optimization + 12-Agent LangGraph + 20 MCP Tools + ML Prediction + A* + Yen's K-Shortest Paths

NEXT
Expand MCP tool ecosystem
Connect additional enterprise systems
Increase historical data for ML
Improve fleet-level optimization

FUTURE
Reusable agent capabilities across enterprise domains

MCP becomes the integration layer for extending agent capabilities into new enterprise systems.

POTENTIAL APPLICATIONS

SUPPLY CHAIN
Vendor Selection & Procurement Optimization

MANUFACTURING
Production Scheduling & Resource Allocation

ENTERPRISE / GOVERNMENT
Compliance & Risk Evaluation

LOGISTICS
Fleet & Network Optimization

One decision intelligence layer.
Many enterprise domains.